11. Sequences and Series

11.4. Comparison Tests for Series

11.5 Learning Objectives

- Determine when it is appropriate to use a comparison test
- Determine an appropriate series to compare a given series to
- Know when and how to use direct comparison and limit comparison tests.

11.4.2. Limit Comparison Test

Theorem 11.4.1 **The Limit Comparison Test**

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms. If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

where c is a finite positive number ($c > 0$), then either both seris converge, or both series diverge.

Example 11.4.1

Does the series $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ converge or diverge?

Solution

Always checking the divergence test at the start, we can confirm the terms of the series go to zero, $\lim_{n\to\infty} \frac{1}{2^n-1} = 0$ so the divergence test is inconclusive.

Then, we can see that $\frac{1}{2^n-1}$ is fairly similar to $\frac{1}{2^n}$ which we know converges because $\sum_{n=1}^{\infty} \frac{1}{2^n} = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$ is a geometric series with $r = \frac{1}{2} < 1$.

Next, if we want to use direct comparison test to show convergence, then we would need

$$\frac{1}{2^n - 1} < \frac{1}{2^n}$$

$$\text{in other words} \quad 2^n - 1 > 2^n$$

which is false! So we **cannot** use direct comparison test.

We can use limit comparison test, however: let $b_n = \frac{1}{2^n}$ and $a_n = \frac{1}{2^n-1}$ then

$$\begin{aligned}\lim_{n\to\infty}\frac{a_n}{b_n} &= \lim_{n\to\infty}\frac{\frac{1}{2^n-1}}{\frac{1}{2^n}}\\ &= \lim_{n\to\infty}\frac{2^n}{2^n-1}\\ &= \lim_{n\to\infty}\frac{1}{1-\frac{1}{2^n}}\\ &= \frac{1}{1-0}\\ &= 1\end{aligned}$$

since $1 > 0$, then **by the limit comparison test** the series $\sum_{n=1}^{\infty}\frac{1}{2^n-1}$ converges.

Example 11.4.2

Does the series $\displaystyle\sum_{n=1}^{\infty}\frac{2n^2+3n}{\sqrt{5+n^5}}$ converge or diverge?

Solution

We start by deciding which test to use:

- divergence test is inconclusive since $\lim_{n\to\infty}\frac{2n^2+3n}{\sqrt{5+n^5}} = 0$ (exercise: work out this limit to show this step and convince yourself)
- this is not something like a telescoping series where the partial sums are easy to simplify and find the limit of, so we skip that idea
- this is not a geometric series, like $\sum_{n=1}^{\infty} ar^n$, so geometric series test does not apply
- this is not a p-series, like $\sum_{n=1}^{\infty}\frac{1}{n^p}$, so p-series test does not apply
- this is not something we really want to integrate, so we will avoid this test for now
- this is a fairly complicated expression to have as the terms in the series, if we could "ignore" some of the terms and simplify it would actually be sort of like a p-series... so let's compare this to something a little simpler.

In order to figure out which series to compare this to, we want a series that grows at the same rate as the given series. Comparing the degrees of the numberator and denominator, we can compare this series to $\sum_{n=1}^{\infty}\frac{n^2}{\sqrt{n^5}} = \sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$, which diverges by the p series test, since $p = \frac{0}{5} < 1$. We can either try using direct comparison test (we would have to show that $\frac{2n^2+3n}{\sqrt{5+n^5}} \geq \frac{n^2}{\sqrt{n^5}}$, which doesn't sound easy) or the limit comparison test.

Using limit comparison test, let $a_n = \frac{2n^2+3n}{\sqrt{5+n^5}}$ and $b_n = \frac{n^2}{\sqrt{n^5}}$. Note, a_n and b_n have positive terms since everything involved is addition and powers of n. Now that we checked the conditions for limit comparison test, let's compute the limit

$$\begin{aligned}
\lim_{n\to\infty}\frac{a_n}{b_n} &= \lim_{n\to\infty}\frac{\frac{2n^2+3n}{\sqrt{5+n^5}}}{\frac{n^2}{\sqrt{n^5}}} \\
&= \lim_{n\to\infty}\frac{2n^2+3n}{\sqrt{5+n^5}}\left(\frac{\sqrt{n^5}}{n^2}\right) \\
&= \lim_{n\to\infty}\frac{2n^2+3n}{\sqrt{5+n^5}}\left(\frac{\frac{1}{n^2}}{\frac{1}{\sqrt{n^5}}}\right) \\
&= \lim_{n\to\infty}\frac{2+\frac{3}{n}}{\sqrt{\frac{5}{n^5}+1}} \\
&= \frac{2+\lim_{n\to\infty}\frac{3}{n}}{\sqrt{\lim_{n\to\infty}\frac{5}{n^5}+1}} \\
&= \frac{2}{\sqrt{1}} \\
&= 2
\end{aligned}$$

Since $2 > 0$, then we know the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ either both diverge or both converge. Since $\sum_{n=1}^{\infty} b_n$ diverges with p-series test, then $\sum_{n=1}^{\infty} a_n$ also diverges by the limit comparison test.

Remark

This is maybe a lot to write, and some of it is explanations of things that might just be happening in your head. A minimally complete solution is reproduced below, anything less than this is probably losing points to something.

Solution

the student starts by deciding to use limt comparison test, and begins the solution there.

Let $a_n = \frac{2n^2+3n}{\sqrt{5+n^5}}$ and $b_n = \frac{n^2}{\sqrt{n^5}}$. Note, a_n and b_n have positive terms since everything involved is addition and powers of n. Then,

$$\begin{aligned}
\lim_{n\to\infty}\frac{a_n}{b_n} &= \lim_{n\to\infty}\frac{\frac{2n^2+3n}{\sqrt{5+n^5}}}{\frac{n^2}{\sqrt{n^5}}} \\
&= \lim_{n\to\infty}\frac{2n^2+3n}{\sqrt{5+n^5}}\left(\frac{\sqrt{n^5}}{n^2}\right) \\
&= \lim_{n\to\infty}\frac{2n^2+3n}{\sqrt{5+n^5}}\left(\frac{\frac{1}{n^2}}{\frac{1}{\sqrt{n^5}}}\right) \\
&= \lim_{n\to\infty}\frac{2+\frac{3}{n}}{\sqrt{\frac{5}{n^5}+1}} \\
&= \frac{2+\lim_{n\to\infty}\frac{3}{n}}{\sqrt{\lim_{n\to\infty}\frac{5}{n^5}+1}} \\
&= 2
\end{aligned}$$

Since $2>0$, then we know the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ either both diverge or both converge. Since $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty}\frac{n^2}{\sqrt{n^5}} = \sum_{n=1}^{\infty}\frac{1}{n^{\frac{1}{2}}}$ diverges with p-series test ($p=\frac{1}{2}<1$), then $\sum_{n=1}^{\infty} a_n$ also diverges by the limit comparison test.

Remark

Notice the solution above

- defines a_n and b_n for their write up at the start
- checks the requirements for limit comparison test (that the terms in both series are positive) with justification
- shows full work for the limit required
- explains why the series $\sum_{n=1}^{\infty} b_n$ diverges
- writes the conclusion about $\sum_{n=1}^{\infty} a_n$ diverging as well with the words "by limit comparison test" and they have used complete sentences.

Every one of these points is needed for a complete solution.

Example 11.4.3

Does the series $\displaystyle\sum_{n=1}^{\infty}\left(1+\frac{3}{n^2}\right)^4\left(\frac{1}{3^n}\right)$ converge or diverge?

Solution

We go through the available tests as we have done before, and settle on limit comparison for the following reasons.

This series is a bit more complex, but thinking about the terms as n gets large, the $\left(1+\frac{3}{n^2}\right)^4$ part goes to just $1^4=1$. The rest is like a geometric series. This indicates we should compare to a geometric series, and because it is more complex we maybe don't want to have to check if $a_n > b_n$ to use direct comparison test, so we will us limit comparison test.

Then, let $a_n = \left(1+\frac{3}{n^2}\right)^4\left(\frac{1}{3^n}\right)$ and $b_n = \frac{1}{3^n}$. Then,

$$\begin{aligned}\lim_{n\to\infty}\frac{a_n}{b_n} &= \lim_{n\to\infty}\frac{\left(1+\frac{3}{n^2}\right)^4\left(\frac{1}{3^n}\right)}{\frac{1}{3^n}}\\ &= \lim_{n\to\infty}\left(1+\frac{3}{n^2}\right)^4\\ &= 1 > 0\end{aligned}$$

Since $\sum_{n=1}^{\infty}\frac{1}{3^n} = \sum_{n=1}^{\infty}\left(\frac{1}{3}\right)^n$ is a geometric series with $r=\frac{1}{3}<1$, it converges. So by the limit comparison test, $\displaystyle\sum_{n=1}^{\infty}\left(1+\frac{3}{n^2}\right)^4\left(\frac{1}{3^n}\right)$ converges as well.

When is the Limit Comparison Test Not Applicable?

The Limit Comparison Test works really well when there is a series that is known to converge or diverge that has the "same" rate of growth as $n\to\infty$.

This makes it particularly useful for algebraic functions as the underlying sequence. For example both

$$\sum_{n=1}^{\infty}\frac{\sqrt[3]{n^3-2n}}{n^4+n^3} \quad \text{or} \quad \sum_{n=1}^{\infty}\frac{3n+1}{n^2+2}$$

are both good series to try limit comparison test on.

However, some series we cannot easily compare with a series we know converges or diverges. For example,

$$\sum_{n=1}^{\infty}\frac{1}{n\ln(n)}$$

If we tried to use limit comparison test to compare this to $\sum_{n=1}^{\infty}\frac{1}{n}$, it would be inconclusive!

$$\lim_{n\to\infty}\frac{\frac{1}{n\ln(n)}}{\frac{1}{n}} = \lim_{n\to\infty}\frac{1}{n\ln(n)}\cdot n = \lim_{n\to\infty}\frac{1}{\ln(n)} = 0$$

and since the limit is 0, the limit comparison test is inconclusive.

11.5b Section Summary:

- We can now use limit comparison test and can tell when limit comparison test is inconclusive.